Financial Statement Fraud Detection Using Large Language Models

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***Abstract*—We develop a new fraud detection model incorporating recent technological advances in Large Language Models such as LLaMA2, LLaMA3, and GPT-4o. We demonstrate that combining deep learning knowledge with domain knowledge benefits fraud detection. We differ from previous research by using large language models to analyze financial, textual, and linguistic data from annual 10-k financial statements.**

***Index Terms*—Fraud Detection, Large Language Models, Machine Learning**

I. Introduction

Fraud is a global issue that spans various industries, and when left unchecked, it can cause serious harm to stakeholders, employees, and other affiliates. Financial fraudulent activities worldwide have amounted to $5.127 trillion, with most of the increase in the past ten years [7]. However, with the rise of artificial intelligence usage in businesses in the past five years, that number is almost certain to rise further and cause more damage. Financial statement fraud, however, is challenging to detect and prevent. Typically, there are 40-50 cases a year and even those that are detected end up being detected years after the fraud had been committed [5]. Standard machine learning models have proven to have good accuracy but need to catch up in correctly predicting the right firms to be fraudulent. With recent technological advances in large language models, more effective methods would be invaluable to regulators, auditors, and investors.

Our study aims to develop a new dataset and new financial statement fraud detection model using available information from the SEC financial statements and the Accounting and Auditing Enforcement Releases (AAERs). We differ from Bao et al. in that we do not limit ourselves to only financial data. We will model our model after Chen, Kelly, and Xiu [2]. They leverage the LLMs such as GPT and LLaMA to analyze all news data to predict stock returns. In the scope of this study, there is little existing literature on using LLMs to analyze financial text, voice, and facial expression data. We provide an implementation using LLMs to improve on existing machine learning results. Similarly to Bao et al., we seek to explain fraud out-of-sample, essentially making this a prediction problem, rather than the standard causal inference problem such as in Dechow et al. [4].

We use five types of fraud prediction models from previous literature as benchmarks: logistic regression, random forest, SVM, XGBoost, and RUSBoost. We use similar performance evaluation metrics as Bao et al. [10] to assess the performance of these models. These will be discussed in section IV of the paper. First, we use the area under the Receiver Operating Characteristics (ROC) curve (AUC). We also use the Normalized Discounted Cumulative Gain at the position k (NDCGk). Thirdly, we modify our measures of sensitivity and precision to reflect the top 1% of predicted fraudulent companies probability.

An exciting question is whether adding more features to our models will improve the performance. Once the LLM is implemented, we must re-evaluate existing machine learning models with the new data. We will also perform a more in-depth analysis of the most important financial variables and ratios based on more current literature. With only 28 financial variables and 14 financial ratios, we do not see the performance increasing significantly unless more information is given about the companies (i.e., MD&A, earnings call data, etc.). Our study will join an extremely small area of financial literature incorporating LLMs to predict financial statement fraud out of sample. Studies like Chen, Kelly, Xiu [2] have paved the way for using LLMs in financial research. Some of the preliminary results raise the possibility that the LLMs could further improve the precision of machine learning models in fraud detection in financial statements.

II. Background

One of the most common accounting principles is the fraud triangle, proposed by Cressey in the 1970s. It consists of three components: perceived financial need, opportunity, and rationalization [3]. Most companies expect significant returns while putting in the minimum effort to maximize their scarce resources. Fraudulent activities are often only discovered after whistleblower tips to the SEC. While fraudulent activities may seem one-dimensional on the surface, there are many indirect costs associated with fraud, including loss of reputation, productivity loss, and irreversible financial damage to employees of the firm that is committing fraud.

III. Methodology

We use several machine learning methods outlined in Bao et al. [10] and choose other common methods to experiment on fraud datasets. We adjust the hyper-parameters to tune the model performance. We use the public dataset from Bao et al. [10] and conduct our baseline analysis on their data. For the initial testing year 2003, we used 1991-2001 as the training set and 2002 as a validation set to tune our models. For each additional year after 2003 up until our last sample test year of 2008, we increase the training and validation sets by one year to increase model performance. For example, for the test year 2008, we use 1991-2006 as the training set, and validate on 2007. Dyck, Morse, and Zingales [5] find it takes, on average, two years for fraudulent activities to be discovered, hence the two-year gap between the training set and test set. Additionally, we set the $random\_state$ to be 42 for ease of recreatibility. We also normalize the features for easier interpretability. Fig. 1 and 2 shows the overall methodology for our experiment.

A. Fine Tuning the LLM

1) Loading the Model: We load the LLaMA2 model using the HuggingFace `AutoModelForCausalLM` with the specified BitsAndBytes configuration. The model is loaded with `quantization_config` set to the BitsAndBytes configuration and `device_map` set to "auto" to automatically allocate the model across available GPUs. The tokenizer is loaded from the Meta LLaMA2 official HuggingFace repository, ensuring compatibility with the model. We set the tokenizer's padding token to be the end-of-sequence (EOS) token to handle variable-length inputs during training.

2) Preprocessing: Preprocessing involves several steps ensuring that the model receives all necessary information to complete the task:

1) **Formatting the Dataset**: Each sample is formatted using the defined prompt structure. The input text is formatted to include an instruction, the input context, and a placeholder for the response. The format is designed to be clear and consistent, ensuring that the model can understand and process the input correctly. The format includes the following sections:
   - **Introduction Blurb**: Provides context that an instruction follows.
   - **Instruction Key**: Specifies the task the model should perform.
   - **Input Key**: Contains the input text to be analyzed.
   - **Response Key**: Placeholder for the model's response.
   - **End Key**: Marks the end of the formatted text.
2) **Tokenization**: The formatted text is tokenized using the LLaMA2 tokenizer, ensuring that each tokenized sample fits within the maximum token length.
3) **Splitting Long Texts**: Long input texts are split into smaller chunks that fit within the token length limit while retaining the instruction and response placeholders in each chunk. We also use overlapping tokens to ensure continuity between the chunks.
4) **Batch Processing**: The tokenized texts are processed in batches to improve efficiency due to our hardware constraints.
5) **Filtering**: Samples that exceed the token length after splitting are truncated, ensuring that all samples fit within the specified limit. In our case, we should not have any samples being filtered out due to splitting, in order to maintain all information.

This preprocessing ensures that the dataset is ready for training, with each sample correctly formatted and tokenized.

3) PEFT Configuration: We use the Parameter-Efficient Fine-Tuning (PEFT) approach with Low-Rank Adaptation (LoRA) to fine-tune the model efficiently. The PEFT configuration includes:

- **LoRA Rank (`lora_r`)**: Determines the size of the low-rank decomposition.
- **LoRA Alpha (`lora_alpha`)**: Scaling factor for the adaptation.
- **LoRA Dropout (`lora_dropout`)**: Dropout rate to regularize the adaptation process.
- **Bias Term Inclusion**: Option to include bias terms in the adapted layers.

This configuration allows for efficient fine-tuning by adapting a subset of the model parameters, reducing computational overhead while maintaining performance.

4) Parameters: We define the following function $fine\_tune$ with various parameters:

```
fine\_tune(model, tokenizer, preprocessed\_dataset,
    lora_r, lora_alpha, lora_dropout, bias,
    task_type, per_device_train_batch_size,
    gradient_accumulation_steps, warmup_steps,
    num_train_epochs, learning_rate, fp16,
    logging_steps, output_dir, optim)
```

Listing 1. Fine Tuning Function

B. Testing Fine-Tuned LLaMA2 Model

After completion of fine tuning, we use our new model to run a simple inference task on the test dataset, which consists of the firms in year 2004. Our goal is to expand the test set further in the future, but for now, we only test year 2004. We take a sample of four firms, two positive and two negative, and input it into the fine-tuned model using the same preprocessing methods used when we fine tuned the model.

IV. Data

A. AAER Dataset

We use the public dataset compiled from Detecting Accounting Fraud in Publicly Traded U.S. Firms Using a Machine Learning Approach. We will refer to this dataset as the JAR dataset in the rest of the paper. They use financial data from Compustat to create the dataset. The dataset consists of two unique identifier variables for each firm: $fyear$ and $gvkey$. The $gvkey$, or Global Company Key, is a six-digit code assigned to each firm in the Compustat database. For the fraud label, they track the case number from the SEC AAER database, which contains a list of all firms that have been audited or investigated for fraud. This is represented by p_aaer in the public dataset, and for every instance of a case number, the label we are predicting, $misstate$, is equal to one. The main features of the public dataset consist of a pairing of 28 financial variables with 14 financial ratios. As the paper we are basing our study on is from 2019, the dataset only consists of cases up until 2018 and needs to be completed. However, we only used the 1991-2008 sample period for our initial testing. The sample began in 1991 due to the stricter enforcement of accounting fraud. The sample ends in 2008 due to the reduction of accounting fraud enforcement due to the higher probability of accounting fraud cases going undetected post-2008 [9].

B. SEC 10-K Dataset

We use the dataset compiled by [8] with annual 10-K filings from the years 1993-2020. For our study, we use a sample set of years 1993-2008 to match the AAER Dataset. We select ITEM 7 from the 10-K annual report, known better as the Management's Discussion and Analysis. This section is commonly used for sentiment analysis and any other forms of textual analysis. We choose to use it for fraud detection because of the complex nature of large language models and their ability to detect any patterns within the texts. Our full dataset consists of nearly 225,000 entries. We use the SEC 10-K Dataset to merge with the AAER dataset and create a training dataset using matched companies and those only in the years 1993-2003. We find that there are around 35,000 matched firms from the years 1993-2003. For testing, we match the years we test on from 2004 and on, and find that around 90% of the firms in the AAER dataset are present in the SEC 10-K dataset.

C. Data Distribution

The total number of fraud instances in the JAR dataset is 964, and the total number of non-fraudulent instances is 145,081, summing up to 146,045 entries total. We also plot the distribution of the number of fraud cases per year and find that it follows a relatively normal distribution with a peak in fraud cases in the 2000-2003 range. When we look at the unique number of fraud cases per year, we find that the shape of the distribution of fraud cases per year is similar, but there are significantly fewer cases in the peak years (2000-2003). Table I displays the most common offenders within the dataset. Figure III shows the number of fraud cases per year with the number of total fraud cases overlapping the number of unique fraud cases per year.

D. Features

We use a list of raw financial variables selected by Bao et al. [10] based on Cecchini et al. [1] and Dechow et al. [4], compiled from the Compustat database. The final dataset from Bao et al., which we use in this study for a baseline, contains 28 financial variables and 14 financial ratios.

V. Performance Evaluation Metrics

Due to the imbalance of our dataset, running standard machine learning methods will result in many false positives. Similarly, if we consider the reality, auditors cannot and will not audit every company. That would take too many resources, and even then, they still may not be accurate in detecting fraud. Therefore, we use the AUC, Precision, Sensitivity, and NCDG@k as defined below from Bao et al.

VI. Machine Learning Results

The preliminary results of our study, as shown in Tables IV-VII, provide an in-depth comparison of various machine learning methods applied to financial prediction tasks using different sets of input variables. By comparing methods across different input sets, we observe notable trends and patterns that shed light on the effectiveness of these techniques and the impact of additional financial ratios. The evaluation metrics considered are AUC (Area Under the Curve), NDCG@k (Normalized Discounted Cumulative Gain at rank k), Sensitivity, and Precision. We see significant enhancements in performance metrics:

- Random Forest emerges as the top performer with an AUC of 0.798 and superior sensitivity and precision. This indicates its robustness in capturing complex patterns and interactions within the financial data, making it a highly effective method.
- XGBoost also demonstrates strong performance, though slightly trailing behind Random Forest. Its notable AUC and precision reflect its ability to handle large datasets with high dimensionality effectively

A. Change in Input

When additional financial ratios are incorporated (Table VI), the performance of the models exhibits mixed results:

- Random Forest maintains a high AUC but experiences reductions in other metrics. This indicates that while the additional ratios provide new information, they might also introduce noise or redundancy, slightly hindering the models sensitivity and precision.
- XGBoost, on the other hand, shows a slight improvement in AUC, suggesting it can leverage the extra features to some extent, but like Random Forest, it faces a trade-off with sensitivity and precision.

VII. LLM Results

Our script finds the max token length to be 4096; however, when we try to train our model using the specified token length, the system does not have enough memory. When training with a token length of 4096, we find that the training halts at 2500 steps, whereas at a token length of 1024, we find that the training completes in about three hours. Therefore, we sample the training and test sets to improve efficiency and produce some results. Rather than using the modified performance metrics, we revert back to the standard metrics in the field: accuracy, recall, precision, and F1-score. We find that our fine tuning either has no impact or our testing has some issues, as the performance is no better than a guess, which we referred to as 50/50 in our machine learning models.

VIII. Future Direction

We present an updated visual to Fig. 1 with the further implementation of LLM in our next step in Figure 4.

A. Dataset Expansion

We plan to increase the dataset's scope into the current year of study, 2025. That would bring valuable training data and present an opportunity to test our fraud detection model on previously untested data. This would require lots of manual work by scraping off of the SEC AAERs webpage to update the dataset up until 2024, as the current dataset only goes until 2018. Additionally, we will leverage the Compustat database to fill in the financial variables for firms from 2018-2024. Once this dataset is complete, we hope to re-run the experiment and determine any improvements and hyperparameter tunings that can be made to the models. Below is a description of what new data could be used for:

B. Large-Language Models (LLMs)

We plan to leverage LLMs to give a fraud score when given an input of financial statement text data rather than predicting the fraud itself, and using this as a signal to predict stock returns, volatility, or short interest.

References

[1] Cecchini, M.; H. Aytug; G. J. Koehler; AND P. Pathak. Detecting Management Fraud in Public Companies. Management Science 56 (2010): 114660.

[2] Chen, Yifei and Kelly, Bryan T. and Xiu, Dacheng, Expected Returns and Large Language Models (November 22, 2022). Available at SSRN: https://ssrn.com/abstract=4416687

[3] Cressey, D.R. (1973). Other Peoples Money: A study in the social psychology of embezzlement. Glencoe: Free Press.

[4] Dechow, P. M.;W. Ge; C. R. Larson; and R. G. Sloan. Predicting Material Accounting Misstatements. Contemporary Accounting Research 28 (2011): 1782.

[5] Dyck, A.; A. Morse; AND L. Zingales. Who Blows the Whistle on Corporate Fraud? Journal of Finance LXV (2010): 221353.

[6] Fawcett, T. An Introduction to ROC Analysis. Pattern Recognition Letters 27 (2006): 86174.

[7] J. Gee, M. Button, The Financial Cost of Fraud 2019, Tech. Rep, Crowe, 2019.

[8] Lefteris Loukas, Manos Fergadiotis, Ion Androutsopoulos, and Prodromos Malakasiotis. 2021. EDGAR-CORPUS: Billions of Tokens Make The World Go Round. In Proceedings of the Third Workshop on Economics and Natural Language Processing, pages 1318, Punta Cana, Dominican Republic. Association for Computational Linguistics.

[9] Rakoff, J. S. The Financial Crisis: Why Have No High-Level Executives Been Prosecuted? The New York Review of Books, January 9, 2014.

[10] Yang Bao, Bin Ke, Bin Li, Julia Yu, and Jie Zhang (2020). Detecting Accounting Fraud in Publicly Traded U.S. Firms Using a Machine Learning Approach. Journal of Accounting Research, 58 (1): 199-235.

Appendix

TABLE I
Performance Evaluation Metrics for the Test Period 2003-08
Input: 28 Financial Variables

Method	AUC	NDCG@k	Sensitivity	Precision
(1) Logit	0.640	0.027	3.345%	2.845%
(2) Random Forest	0.798	0.143	15.192%	11.706%
(3) RUSBoost	0.770	0.051	4.963%	5.099%
(4) SVM	0.638	0.024	3.057%	2.562%
(5) XGBoost	0.760	0.099	9.634%	7.687%

TABLE II
Performance Evaluation Metrics for the Test Period 2003-08
Input: 28 Financial Variables + 14 Financial Ratios

Method	AUC	NDCG@k	Sensitivity	Precision
(1) Logit	0.703	0.022	3.103%	2.562%
(2) Random Forest	0.790	0.084	7.689%	6.007%
(3) RUSBoost	0.751	0.029	2.996%	2.835%
(4) SVM	0.708	0.027	3.084%	2.285%
(5) XGBoost	0.713	0.070	8.121%	6.546%

TABLE III
Percentage Change From Table IV to Table V

Method	AUC	NDCG@k	Sensitivity	Precision
(1) Logit	9.84%	-18.52%	-9.95%	-7.24%
(2) Random Forest	-1.00%	-41.26%	-48.68%	-49.39%
(3) RUSBoost	-2.47%	-43.14%	-44.40%	-39.63%
(4) SVM	10.97%	12.50%	-10.81%	0.88%
(5) XGBoost	-6.18%	-29.29%	-14.84%	-15.71%

TABLE IV
Reported Evaluation Metrics for the Test Period 2003-08
Input: 28 Financial Variables

Method	AUC	NDCG@k	Sensitivity	Precision
(1) Logit	0.690	0.006	0.73%	0.85%
(2) RUSBoost	0.725	0.049	4.88%	4.48%
(3) SVM	0.680	0.016	1.69%	1.90%

TABLE V
Percentage Change From Table VII to Table IV

Method	AUC	NDCG@k	Sensitivity	Precision
(1) Logit	-7.25%	350.00%	358.22%	234.71%
(2) RUSBoost	6.21%	4.08%	1.70%	13.82%
(3) SVM	-6.18%	50.00%	80.89%	34.84%

TABLE VI
Model Performance Metrics

Metric	Value
Accuracy	0.5000
Precision	0.5000
Recall	1.0000
F1-score	0.6667

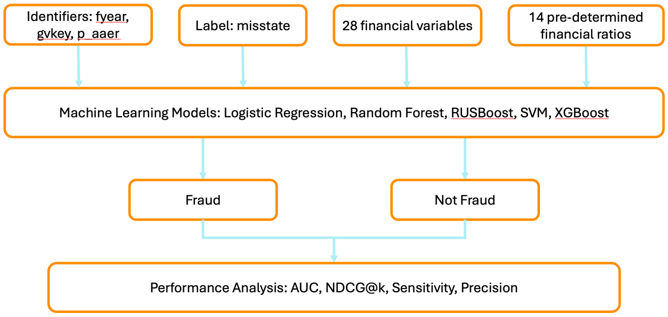

Fig. 1. Figure of Experiment

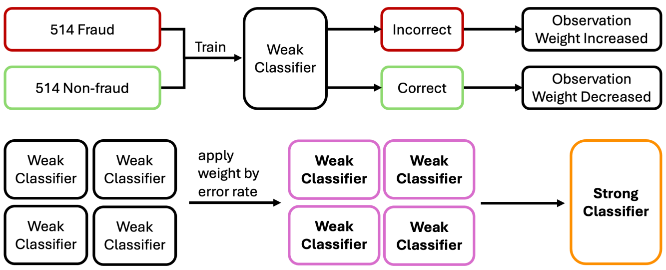

Fig. 2. Depiction of RUSBoost model. The top half repeats itself, and the bottom half takes a weighted average of all the weak classifiers from the top.

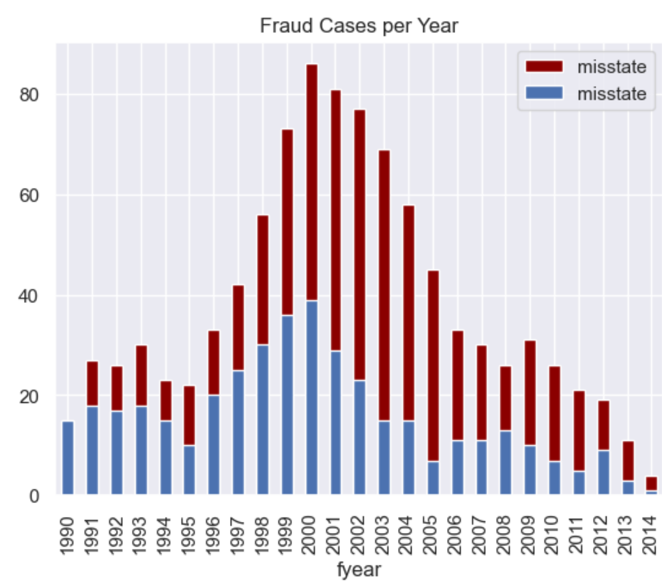

Fig. 3. The blue bars are new cases for every year, and the red bars represent the total number of fraud cases, including repeat offenders.

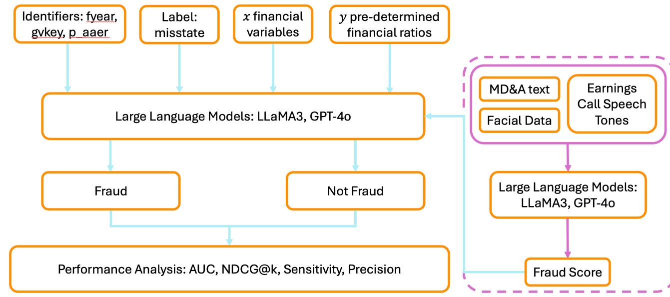

Fig. 4. Figure of Future Experiments